How AdamW and Muon Train a Small Language Model

Next steps (ii) and (iii) of the checkpoint

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Summary. (ii) The update cosine follows the rule; so do stable rank and $\sigma_1/\sigma_{\text{median}}$, but the spectral norm does not. (iii) Weight decay does not explain Muon's speed; without orthogonalization Muon is worse; speed and spectra move together, so the cause is open; after a swap the loss moves before the stable rank.

1 Setup and reproduction

TinyStories-10M on FAS RC Cannon, the checkpoint's settings (AdamW $2 \cdot 10^{-3}$, Muon 0.01, 10,000 steps, data orders 0 to 2); spectral metrics are means over the 36 block matrices. The rerun agrees (Table 1).

Table 1: Rerun (checkpoint in brackets); the last two rows at training loss 1.81.

	AdamW	Muon
final validation loss	1.339 to 1.345 (1.344)	1.308 to 1.310 (1.309)
step reaching AdamW's final validation loss		7250 to 7750 (7500)
stable rank of W	32.7 ± 0.2 (33.0)	86.3 ± 0.5 (86.2)
active MLP units	0.120 ± 0.001 (0.120)	0.222 ± 0.001 (0.222)

2 (ii) Direct result of Muon's updates, or emergent in training?

2.1 Rules without learning

On synthetic gradients (noise, or a rank-4 signal plus noise), each rule updates one matrix for 10,000 steps (Fig. 1).

Update cosine follows the rule: on noise, 0.90 for AdamW and 0.65 to 0.76 for Muon, close to the real runs (0.89, 0.74).

The spectra do not: on noise the stable rank is unchanged; with the rank-4 signal both fall to about 4 (step 1000: AdamW 3.7, Muon 10.2).

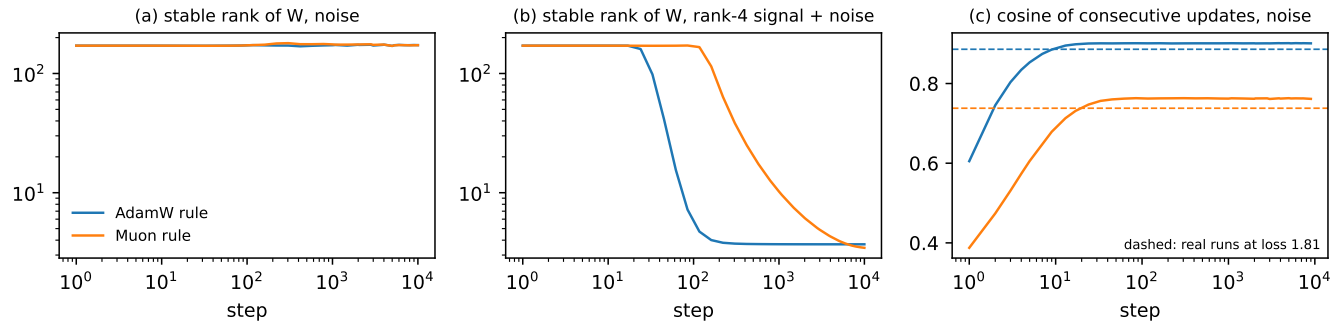


Figure 1: Each rule on synthetic gradients, MLP input matrix.

2.2 Ghosts on real gradients

Ghost: a copy of the initial weights updated by the other rule on the run's own gradients, never fed back (Table 2 and Fig. 2).

The ordering follows the rule: on the same gradients Muon's rule gives the higher stable rank (87 against 40; 75 against 13).

The spectral norm does not: which rule is larger depends on the gradient stream (10.9 against 5.9 to 6.1; 7.5 against 7.7 to 7.8).

Table 2: The 2×2 design at step 10,000; range over data orders 0 to 2.

rule	gradient stream	stable rank	$\sigma_1/\sigma_{\text{median}}$	update cosine
AdamW	AdamW (run)	39.9 to 41.4	5.80 to 6.19	0.89 to 0.90
AdamW	Muon (ghost)	12.6 to 13.0	15.6 to 16.0	0.89
Muon	Muon (run)	75.3 to 75.5	2.89 to 2.90	0.71
Muon	AdamW (ghost)	87.0 to 87.2	2.77 to 2.80	0.70

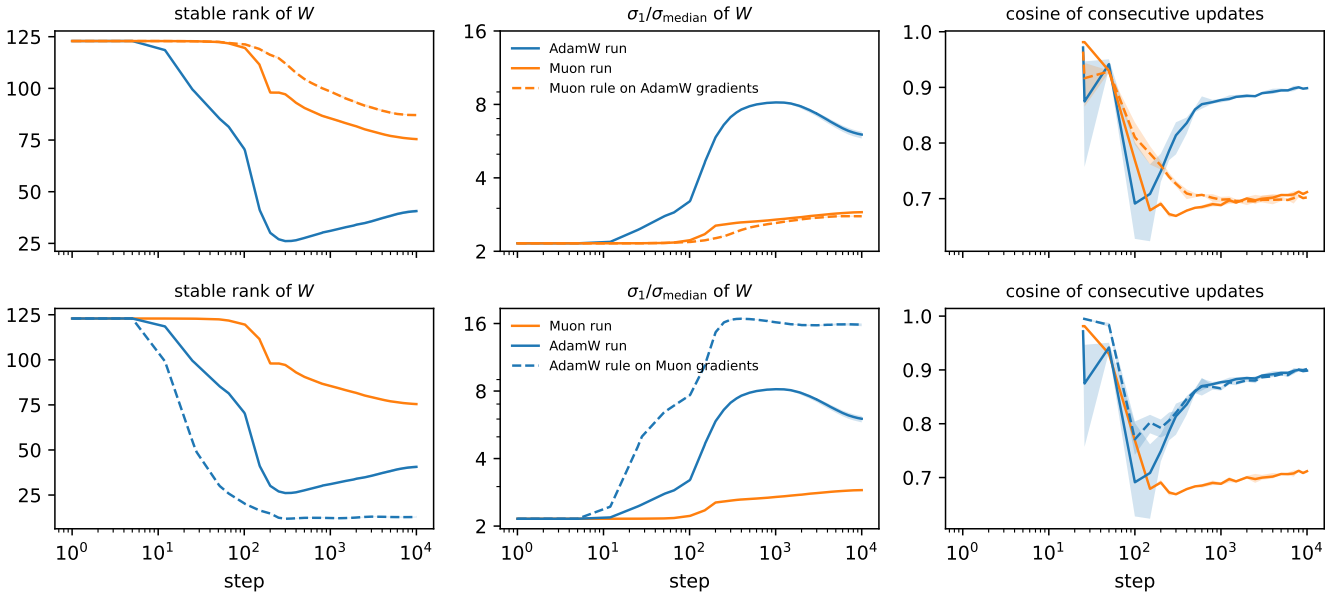


Figure 2: Ghosts (dashed) against runs; mean over data orders 0 to 2.

3 (iii) Do the differences explain Muon's faster training and different path?

3.1 What the speed needs

Weight decay is not the reason (Table 3): Muon has none and is faster; adding or removing it changes final loss by 0.005 to 0.012.

Without orthogonalization (momentum rescaled to an orthogonal update's norm) Muon is worse: worse than AdamW at every learning rate; at 0.01 the two failing orders collapse to stable rank 3.6 and 5.5 (the order that trains: 34.8).

Table 3: Final validation loss at step 10,000.

run	data order		
	0	1	2
AdamW (baseline)	1.339	1.341	1.345
AdamW, weight decay 0	1.346	1.353	1.350
Muon (baseline)	1.308	1.309	1.310
Muon, weight decay 0.02	1.297	1.298	1.304
Muon, no orthogonalization, lr 0.01	3.729	2.943	1.389
Muon, no orthogonalization, lr 0.003	1.454		
Muon, no orthogonalization, lr 0.03	4.078		

3.2 Dose-response

Update singular values raised to p: $p = 0$ is exact orthogonalization, $p = 1$ none (Table 4; Muon, lr 0.01, order 0).

Speed and spectra move together with p, toward AdamW; the cause is open. $p = 0.75$ on orders 0 and 1 is close to AdamW in stable rank and in the step reaching loss 1.81 (33.4 to 33.5 against 32.4 to 32.9; 1379 to 1394 against 1426 to 1469); order 2 stalls and ends at 1.659.

Table 4: Muon with singular values raised to p, data order 0; step and spectra at training loss 1.81.

run	step at loss 1.81	stable rank	$\sigma_1/\sigma_{\text{median}}$	final val. loss
Muon (Newton-Schulz)	846	86.6	2.68	1.308
$p = 0$ (exact SVD)	827	88.2	2.64	1.306
$p = 0.25$	949	63.8	3.39	1.321
$p = 0.5$	1230	41.3	5.14	1.341
$p = 0.75$	1394	33.5	7.61	1.357
AdamW	1469	32.9	7.99	1.339

3.3 Swap timescale

Swap rules at step k (ORCD, order 0, compared only with the swap's own parents; Fig. 3).

From step 1000 on, the loss moves before the stable rank: AdamW to Muon, at +100 steps the loss gap is −0.062 (parents −0.146) while the stable rank has moved 9%.

Exception: Muon to AdamW at step 400, the stable rank has moved 73% by +25, together with the loss spike.

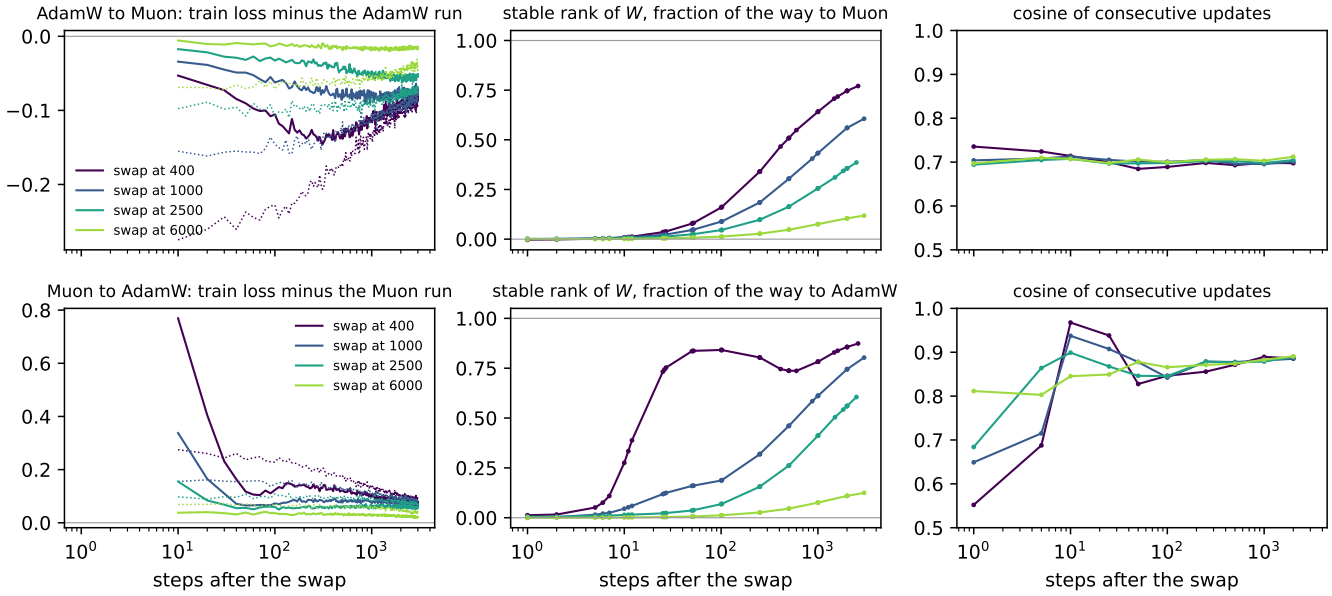


Figure 3: Swaps at steps 400, 1000, 2500 and 6000. Dotted: the parents' loss gap.